Morphological Processing

Week 7 — Session 2: Advanced operators & the cleanup pipeline

Practical Image Processing

Lecture 7.2 ▪ 60–70 min

Recap of Session 1

- **Erosion** (fit) shrinks; **dilation** (hit) grows — set by a **structuring element**.
- **Opening** = erode→dilate removes small bright specks (keeps size).
- **Closing** = dilate→erode fills small gaps/holes (keeps size).

Today

Compose these into **gradient**, **top-hat**, **black-hat**; go **grayscale**; remove lines; and build the DocScan cleanup pipeline + deliverable.

Roadmap

Derived operators

Grayscale morphology

Structuring-element design

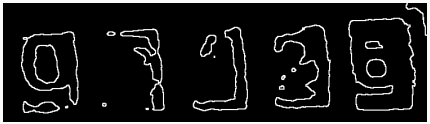
The cleanup pipeline

In-class exercise

Wrap-up & deliverable

Derived operators

Morphological gradient, top-hat, black-hat



gradient = outline

- **Gradient** = dilation − erosion → the object outline (thickness ≈ SE size).
- **Top-hat** = image − opening → small *bright* details on a darker background.
- **Black-hat** = closing − image → small *dark* details on a brighter background.

✔ Good practice

Top-hat is a lighting fix: it isolates the digits from a slowly-varying background.

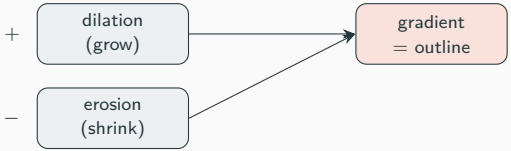
Why top-hat corrects uneven lighting

- Opening with a *large* SE keeps only the smooth background (it erases small bright features).
- Subtracting that background — `image - opening` — leaves the small bright digits, flattened.
- Black-hat is the mirror image: it recovers small dark features sitting on a bright field.

Math

Top-hat$(f) = f - (f \circ B)$; Black-hat$(f) = (f \bullet B) - f$. Choose B larger than the details you want to keep.

Gradient as an edge detector for shapes



- The gradient is a band between the grown and shrunk shape — a clean, closed contour.
- Unlike a Sobel edge, it comes straight from a binary mask and is already one connected ring.

Grayscale morphology

Grayscale morphology

- Morphology also works on grayscale: **erosion** = local *minimum*, **dilation** = local *maximum* under the SE.
- Opening/closing/top-hat all carry over — no thresholding required first.
- Top-hat on grayscale removes a smooth background gradient *before* you threshold.

Key idea

Morphology is not only for binary — on grayscale it is a nonlinear min/max filter that respects shape.

Grayscale erosion — worked example

Grayscale erosion replaces each pixel with the *min* over a 1×3 window; dilation with the *max*.

input	40	90	90	20	90	90	40
erode (min)	40	40	20	20	20	40	40
dilate (max)	90	90	90	90	90	90	90

- The dark dip (20) spreads under erosion; bright plateaus spread under dilation.
- Top-hat here = input − opening would flag that narrow dark notch.

Structuring-element design

Structuring-element shapes

Shape	When	Note
Rectangle	general cleanup	fast, blocky effect
Ellipse	round objects	smoother, more natural
Cross	thin connectors	minimal, gentle
Line (h/v)	directional	remove/keep horizontal or vertical strokes

Key idea

A line SE can remove table lines while keeping text — the shape of the probe is a design choice.

Shape & size both matter

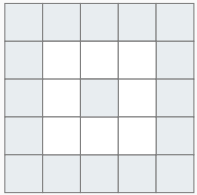
- A larger SE = a stronger effect (removes bigger specks / fills bigger gaps) but more distortion.
- Shape matters: a vertical SE affects vertical strokes differently than a horizontal one.
- Rule: use the *smallest* SE that fixes the defect — bigger erases real detail.

✕ Pitfall

A too-large SE turns your digits into unrecognizable blobs. Tune the size on real data, not on a toy image.

Boundary extraction & hole filling

- **Boundary** = image − erosion (a thin inner outline; cf. the gradient which straddles the edge).
- **Hole filling**: fill enclosed dark regions inside a shape (a solid digit from a hollow one).
- Both compose from erosion/dilation — a small toolkit that builds many effects.



hollow → filled

Skeletonization, thinning & hit-or-miss

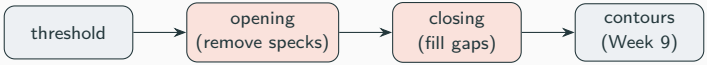
- **Skeleton**: repeatedly thin a shape to a 1-pixel "spine" — its topological essence.
- Useful for character/stroke analysis and measuring connectivity.
- **Hit-or-miss** finds exact local patterns (corners, endpoints, junctions) with a foreground *and* a background SE.

Note

Skeletons are fragile to noise — clean with opening *first*, or the spine sprouts spurious branches.

The cleanup pipeline

A morphology cleanup pipeline



- Typical DocScan order: threshold → open (kill noise) → close (repair digits) → contours.
- Order matters: opening before closing avoids fusing a speck into a digit.

Removing lines, keeping text

- A form has table/underline strokes you want gone but the text kept.
- Use a long *horizontal* line SE with opening to isolate horizontal lines, then subtract them.
- A directional SE is a precise tool — it operates only on structures of a chosen orientation and length.

✓ Good practice

For a scanned form (a DocScan case), this cleanly separates rules/boxes from the handwritten content.

Line removal in code

```
# isolate long horizontal rules, then subtract them
hks = cv2.getStructuringElement(cv2.MORPH_RECT, (25, 1))
lines = cv2.morphologyEx(binary, cv2.MORPH_OPEN, hks)
text_only = cv2.subtract(binary, lines)
```

- The 25×1 SE survives only along long horizontal runs — the table rules.
- Subtracting them leaves the digits; swap to $(1, 25)$ for vertical rules.

Order of operations matters

- **Open-then-close** vs **close-then-open** give *different* results — the operations do not commute.
- Open first to kill specks, *then* close: a leftover speck can otherwise be sealed into a digit.
- Top-hat/black-hat and gradient are also non-commuting compositions — always state the order.

✕ Pitfall

Closing before opening can fuse a noise speck onto a stroke, creating a fake serif the OCR misreads.

Choosing the SE size for line removal

- The horizontal SE length must exceed the *thickest text stroke* but stay below the *shortest rule* you want gone.
- Too short: text survives into the "lines" mask and gets subtracted away. Too long: short rules escape.
- Practical recipe: set length $\approx 1.5\times$ the character width; verify on a real scan.

Math

A $(25, 1)$ SE keeps runs $\geq$ 25px wide horizontally — tune 25 to your scan resolution.

Granulometry — measuring size by morphology

- Open with SEs of increasing size and record how much foreground area disappears at each step.
- The drop at size s counts objects roughly of size s — a size *distribution* without segmentation.
- A quick way to ask "how big are my specks?" before fixing the SE for opening.

Note

Granulometry is morphology used as a *measurement* tool, not just a cleanup tool.

Morphology vs blurring

- Blurring averages (softens everything); morphology reasons about *shape* (keeps edges hard).
- To remove specks from a *binary* mask, opening beats a blur (which would gray it out).
- Use blur on grayscale/noise; use morphology on binary shapes.

Key idea

Morphology is the right tool once you have a binary mask — it is shape surgery, not smoothing.

In-class exercise

In-class exercise (10 min)

Design the operator

Match each goal to a *derived* operator and an SE, then say what you would subtract.

1. Draw a one-pixel outline around every digit.
2. Flatten a photo whose left side is darker than its right, before thresholding.
3. Delete the underline printed beneath the meter reading, keeping the digits.
4. Recover a faint dark serial number printed on a bright label.

Debrief

- (1) **Gradient** (dilate − erode) with a 3×3 SE.
- (2) **Top-hat** with a *large* SE, then threshold the flattened result.
- (3) **Opening** with a long horizontal line SE, then `subtract`.
- (4) **Black-hat** with an SE larger than the characters.

Key idea

Every derived operator is just erosion/dilation composed and subtracted — name the base, pick the SE, choose the sign.

Wrap-up & deliverable

Derived operators — one-glance summary

Operator	Definition	Use
Gradient	dilation − erosion	object outline
Top-hat	image − opening	small bright details, lighting fix
Black-hat	closing − image	small dark details
Boundary	image − erosion	inner one-pixel edge

Key idea

Every row is erosion/dilation composed and subtracted. Learn the base pair and the rest are bookkeeping.

All the operators in one call

```
k     = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (3, 3))
grad  = cv2.morphologyEx(binary, cv2.MORPH_GRADIENT, k)   # outline
top   = cv2.morphologyEx(gray,   cv2.MORPH_TOPHAT,   big_k) # lighting fix
black = cv2.morphologyEx(gray,   cv2.MORPH_BLACKHAT, big_k) # dark details
```

`morphologyEx` covers open/close/gradient/top-hat/black-hat with one flag.

Grayscale vs binary — which pipeline?

You have	Do morphology…	Then
a clean scene, even light	after thresholding (binary)	open → close
uneven lighting	*before* thresholding (top-hat on gray)	then threshold
faint dark marks	black-hat on gray	then threshold

- Grayscale morphology *prepares* the image; binary morphology *cleans* the mask.
- For a hard DocScan photo: top-hat, threshold, open, close — in that order.

What you should now be able to do

- Apply gradient / top-hat / black-hat, and correct uneven lighting with top-hat.
- Explain grayscale morphology as a min/max filter; trace it on a row of values.
- Design a directional SE to remove lines while keeping text.
- Build a threshold $\rightarrow$ open $\rightarrow$ close cleanup pipeline for the digits.

This week's deliverable

Due before Week 8 (see `project/`)

On a thresholded meter/plate (from Week 2):

1. Demonstrate erosion, dilation, opening, closing; explain each effect.
2. Use opening to remove speckle noise and closing to repair broken digits.
3. Apply top-hat to a photo with uneven lighting; compare to before.
4. Use a directional line SE to strip a printed rule/underline while keeping the digits.
5. DocScan milestone: produce a clean binary where each digit is a solid, separable blob.

Threshold makes it binary; morphology makes it clean.

Compose erosion and dilation — gradient, top-hat, and a cleanup pipeline for the digits.